

Industrial Anomaly Detection

MVTec Bottle Category

Technical Analysis Document

Introduction to Computer Vision — Final Project

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Pipeline: Preprocessing → Feature Extraction (ResNet18) → Anomaly Scoring (k-NN) →
Post-processing

1. Problem Statement

Automated visual inspection is a critical component of modern manufacturing pipelines. Human inspectors are slow, expensive, and subject to fatigue-induced errors — particularly in high-throughput production environments where hundreds of items must be evaluated per minute. Computer Vision offers a scalable alternative: a camera-based system that can evaluate every product consistently and in real time.

The central challenge is that defective products are rare by design. A well-functioning production line produces very few defects, making it impractical to collect a balanced labeled dataset for supervised classification. Furthermore, the types of defects that occur in the future may differ from those observed historically, making a closed-set classifier brittle.

This project adopts a **one-class learning** strategy: the model is trained exclusively on defect-free images and learns a compact representation of normal appearance. At test time, any image whose features deviate significantly from the learned normality is flagged as anomalous. This approach requires no defective samples during training and generalizes naturally to previously unseen defect types.

The dataset used is the **MVTec Anomaly Detection dataset** (Bergmann et al., 2019), specifically the *bottle* category. It provides ~209 normal training images, a test set with both normal and defective images across three defect types (broken_large, broken_small, contamination), and pixel-level ground truth masks for all defective samples.

2. Methodology

2.1 Preprocessing

All images undergo a standardized preprocessing pipeline before feature extraction. OpenCV loads images in BGR format; they are converted to RGB to match the convention expected by the ResNet18 backbone. Images are then resized from their native 900x900 resolution to 224x224 pixels — the input size ResNet18 was designed for.

Pixel values are rescaled from [0, 255] to [0.0, 1.0] and then normalized using ImageNet channel statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). This normalization is mandatory because ResNet18 was trained on data with this exact statistical distribution; deviating from it degrades feature quality.

During training only, random horizontal flips ($p=0.5$) are applied as data augmentation to improve the coverage of the normal feature space. Color-based augmentations are intentionally avoided, since color consistency is a relevant cue for contamination detection.

2.2 Feature Extraction

A ResNet18 backbone pre-trained on ImageNet is used as a fixed feature extractor, with no fine-tuning on the MVTEC dataset. The network is truncated after layer2, discarding the deeper layers and the classification head. For a 224x224 input, layer2 produces a feature map of shape (128, 28, 28).

This output is interpreted as a 28x28 grid of **patch descriptors**: each of the 784 spatial positions corresponds to a receptive field of approximately 8x8 pixels in the original image, described by a 128-dimensional feature vector encoding local appearance. This patch-level representation enables spatial defect localization in addition to image-level detection.

Layer2 is chosen over deeper layers for two reasons: it preserves sufficient spatial resolution (28x28 vs 14x14 or 7x7) for localization, and its features capture mid-level visual patterns (textures, local structures) that are more relevant to surface defect detection than the high-level semantic features of deeper layers.

2.3 Memory Bank and k-NN Anomaly Scoring

All patch descriptors from all normal training images are collected into a **memory bank** — a complete reference catalogue of normal local appearances. With ~209 training images and 784 patches each, the full memory bank contains approximately 163,000 vectors of dimension 128. For computational efficiency, the memory bank is randomly subsampled to 50,000 vectors.

A NearestNeighbors model (k=9, Euclidean distance, ball_tree algorithm) is fitted on the memory bank. At test time, each of the 784 patch descriptors of a test image is queried against the memory bank. The mean distance to the 9 nearest neighbors is the patch-level anomaly score: low distance indicates a patch that resembles known normal regions; high distance indicates an unusual local appearance.

The image-level anomaly score is the **maximum** patch score across all 784 positions. The maximum operator is preferred over the mean because defects are spatially localized and averaging would dilute the anomaly signal across the many surrounding normal patches.

2.4 Post-processing

The raw 28x28 patch score map is processed into a pixel-level anomaly mask through four sequential steps:

Step	Operation	Purpose
1	Gaussian smoothing (sigma=4)	Remove block artifacts at patch boundaries
2	Bilinear upsampling to 224x224	Restore full image resolution
3	Normalization + 95th percentile threshold	Convert continuous scores to binary mask
4	Morphological closing then opening (5x5 ellipse)	Fill holes, remove isolated noise

3. Experimental Results

3.1 Quantitative Metrics

The system is evaluated at two levels. At the image level, AUROC measures how well the anomaly score separates normal from defective images across all possible thresholds. At the pixel level, IoU and Dice measure the spatial overlap between the predicted binary mask and the ground truth annotation provided by MVTec.

Metric	Value	Notes
Image-level AUROC	0.9937	Near-perfect image classification
Pixel-level IoU	0.2779	Jaccard index on defective images only
Pixel-level Dice	0.4349	F1-score at pixel level

3.2 Discussion

The image-level AUROC of 0.9937 indicates that the system almost perfectly ranks defective images above normal ones. The score distributions are well-separated: normal images cluster around a mean score of 2.621 (std 0.297), while defective images cluster around 4.302 (std 0.672), with minimal overlap.

Pixel-level performance is more moderate. An IoU of 0.2779 and Dice of 0.4349 reflect the known limitation of patch-based methods at coarse resolution: the system correctly identifies the general location of defects but tends to underestimate their spatial extent. The predicted masks are smaller than the ground truth annotations, particularly for large defects such as `broken_large`.

4. Failure Analysis

4.1 Mask Undersegmentation

The most consistent failure mode is undersegmentation: the predicted mask correctly localizes the defect but covers a smaller area than the ground truth. This occurs because the 95th percentile threshold flags only the single most anomalous patch or a small cluster of patches, while the surrounding defective region — which is anomalous but less extreme — falls below the threshold. A lower threshold or a region-growing post-processing step would improve recall at the cost of precision.

4.2 Small Defects (`broken_small`)

Thin fractures that span only one or two patches produce a weak anomaly signal that is further attenuated by Gaussian smoothing. After upsampling, the signal may not reach the 95th percentile threshold. These cases are the most challenging and likely account for the majority of false negatives at the pixel level.

4.3 Boundary Region Artifacts

Patches near the bottle boundary are structurally different from patches in the bottle interior, and may be underrepresented in the memory bank if the bottle occupies slightly different spatial positions across training images. This can produce elevated anomaly scores at the image periphery even for normal test images, reducing specificity for contamination defects located near the edges.

4.4 Per-image Threshold

The 95th percentile threshold is applied independently to each image, meaning every image — including normal ones — produces a non-empty predicted mask. For normal test images this always generates false positive pixels, which inflates the denominator in IoU computation. A globally

calibrated threshold, estimated on a held-out validation set, would address this issue.

5. Ethical Considerations

5.1 Deployment Bias and False Negatives

In a real industrial deployment, false negatives — defective products classified as normal — carry safety implications if the product reaches end consumers. The system should be validated extensively on the specific production line before replacing human inspection entirely. A hybrid approach, where the model flags uncertain cases for human review, is preferable to full automation.

5.2 Distribution Shift

The memory bank is built from a fixed training set. If the appearance of normal products changes over time — due to changes in materials, lighting conditions, or camera calibration — the memory bank becomes stale and false positive rates increase. Periodic retraining or online memory bank updates are necessary for robust long-term deployment.

5.3 Dataset Representativeness

MVTec AD is a controlled research benchmark. Images are captured under consistent laboratory lighting with a fixed camera setup. Real production environments are noisier and more variable. Performance metrics obtained on MVTEC may not transfer directly to deployment settings, and thorough in-situ validation is required before drawing conclusions about real-world effectiveness.

5.4 Privacy

This project processes images of industrial objects and involves no personal data, biometric information, or identifiable individuals. No privacy concerns arise from the dataset or the methodology.

References

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